

SOP-12: Oxidizing Salts and Metal Compounds

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Copper(II) nitrate hydrate, 99.999% trace metals basis	13778-31-9
Zirconium oxide beads (ZrOB015)	1314-23-4

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Flame over circle (oxidizing solid Cat 2); Corrosion (eye damage Cat 1); Exclamation mark (acute toxicity Cat 4); Environment (aquatic toxicity Cat 1)

- Stock chemicals: copper(II) nitrate hydrate (5 g solid) and zirconium oxide bead-beating beads (400 mg). Intermediates and working solutions: copper(II) nitrate stock and working solutions in acidified deionized water or filtered seawater, used for trace-metal copper exposure experiments and as an analytical standard; zirconium oxide beads used dry in bead-beating tubes for tissue homogenization. Wastes: copper-containing solutions and rinses, used beads in lysate, and contaminated debris.
- Copper(II) nitrate is an OXIDIZER (Oxidizing solids Category 2, MyChem Oxidizer Class 1). In contact with combustible material - paper, wood, cloth, absorbent, organic solvents, reducing agents - it can intensify or initiate a fire. It is also strongly corrosive to the eyes and severely toxic to aquatic organisms.

- Copper is acutely toxic to aquatic life at Category 1 for both acute and chronic endpoints. In a marine laboratory this is the controlling hazard: microgram-per-liter concentrations of dissolved copper are lethal to bivalve larvae and many invertebrates. A copper release into a shared seawater system can destroy other researchers' experiments as well as harming the environment.
- Zirconium oxide beads are chemically inert and are not classified as hazardous. Their hazards are physical: a bead-beating tube that fails releases beads and aerosolized lysate at high energy, and beads on the floor are a slip hazard. Their hazard in practice comes from what is in the tube with them (see SOP-05).
- Routes of exposure: inhalation of dust when weighing copper(II) nitrate, skin contact, eye contact, ingestion.
- How exposure might occur: weighing the hygroscopic copper salt, dissolving it, pipetting stock solutions, dosing exposure vessels, handling contaminated glassware, and loading or opening bead-beating tubes.
- Target organs: eyes (serious damage), skin, gastrointestinal tract; systemic copper overload affects the liver. Nitrate can be reduced to nitrite, which affects hemoglobin at high ingested doses.
- Signs and symptoms of exposure: eye pain, redness and possible corneal damage; skin irritation; metallic taste, nausea and vomiting on ingestion; a blue-green stain on skin or surfaces indicates copper contamination.
- Expected by-products: copper(II) nitrate decomposes on strong heating to release nitrogen dioxide, a toxic brown gas. Do not heat it. Contact with strong acid can release nitrogen oxides.
- GHS classification (from MyChem / SDS): Copper(II) nitrate hydrate - Oxidizing solids Category 2; Acute toxicity Category 4 (oral); Serious eye damage/eye irritation Category 1; Skin corrosion/irritation Category 2; Acute aquatic toxicity Category 1; Chronic aquatic toxicity Category 1. MyChem fire class: Oxidizer, Class 1; Corrosive; Other Health Hazards - Including Carcinogens. Zirconium oxide - not classified.
- Occupational exposure limits: copper, dusts and mists (as Cu), 1 mg/m³ WISHA/OSHA PEL 8-hr TWA. Neither of these chemicals is flagged as a particularly hazardous substance in MyChem; PHS Sections 9-11 are optional and are included below as good practice.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- Weighing copper(II) nitrate and preparing stock solutions is performed inside the certified chemical fume hood in FTR 209, with the analytical balance placed inside the hood, on a disposable absorbent liner, with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S. Because this material is an oxidizer, the absorbent liner is a plastic-backed pad used only for the duration of the weighing and is removed and disposed of promptly rather than left in place with oxidizer residue.
- Dilute working solutions may be prepared and dosed at the bench in closed or covered vessels.
- Bead-beating with zirconium oxide beads is performed inside the fume hood in sealed screw-cap or O-ring tubes; tubes are allowed to settle for one minute before opening, inside the hood (see SOP-05).
- Copper solutions and exposure vessels are handled over secondary containment at all times so that no release can reach a floor drain or a seawater line.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Disposable chemically resistant sleeves are worn for weighing copper(II) nitrate. A chemically resistant apron and face shield are required for spill cleanup.
- Because copper(II) nitrate causes serious eye damage and is a skin irritant, additional protective clothing is required where splashes or skin contact are foreseeable.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: 6-8 mil nitrile gloves for weighing copper(II) nitrate and for handling its solutions, changed immediately on any contact. Nitrile is compatible with aqueous copper salt solutions.
- Note that gloves contaminated with an oxidizer should not be left in contact with combustible material; discard them promptly into the designated waste container rather than leaving them on the bench.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Nitrile gloves are worn when loading and unloading bead-beating tubes.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for weighing copper(II) nitrate and for handling its concentrated solutions, because it causes serious eye damage (Category 1). Safety glasses with side shields are acceptable for dilute working solutions and for bead-beating tube handling. A face shield over goggles is required for spill cleanup.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided powder weighing is confined to the certified chemical fume hood as written.
- Copper(II) nitrate POWDER is not permitted to be weighed or transferred outside of the chemical fume hood. Dilute aqueous working solutions may be used at the bench. Dry zirconium oxide beads may be dispensed at the bench; they generate no respirable dust.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.
- Where air-purifying respirators are appropriate, use a full-face respirator with a P100 particulate filter as a backup to engineering controls.

Section 4 - Special Handling and Storage Requirements

- OXIDIZER SEGREGATION is the primary storage control. Copper(II) nitrate must be stored apart from all flammable and combustible materials: solvents (SOP-01), DMSO (SOP-02), chloroform (SOP-03), paper, cardboard, absorbent, wooden shelving, and all reducing agents including 2-mercaptoethanol and TCEP (SOP-09). Store it in a non-combustible secondary container on a metal or coated-metal shelf, not in a cardboard box.
- Do not heat copper(II) nitrate, and do not mix it with strong acids or with organic material.
- Quantity limits: no more than 1 g is weighed in a single operation; stock solutions are prepared at the smallest practical volume.
- AQUATIC RELEASE CONTROL: no copper-containing solution, rinse, or exposure water may enter a sink, floor drain, seawater intake or return, or any container that will be emptied to a drain. All copper-containing liquid is collected as hazardous waste. Exposure vessels are kept in secondary containment and are labeled 'CONTAINS COPPER - NO DRAIN'.
- All containers - the salt bottle, stock tubes, working solution bottles, exposure vessels and waste containers - must be labeled with the full chemical name, concentration, the words 'OXIDIZER' and 'TOXIC TO AQUATIC LIFE - NO DRAIN DISPOSAL', the date and the preparer's initials.
- Unattended operations: copper exposure experiments in closed, labeled, contained vessels may run unattended provided the area is posted with the chemical, the responsible person and a contact number, and the secondary containment is adequate for the full volume.
- Bead-beating tubes are inspected for cracks before use and are not overfilled; a failed tube in a bead beater is treated as a spill of whatever the tube contained.
- Storage: copper(II) nitrate is stored in its original container inside a labeled, non-combustible secondary container in the oxidizers position of the chemical storage cabinet in FTR 209, segregated from flammables, organics and reducing agents. Zirconium oxide beads are stored in their supplier container at room temperature with no special segregation.
- Transport: containers travel in closed, leak-proof secondary containment.
- Clean the chemical fume hood upon completion of tasks: remove and bag the absorbent liner promptly, then wipe the work surface and balance with a damp wipe, then detergent solution, then clean water; wipe dry. A blue-green residue means copper remains - repeat.
- Clean all contaminated surfaces with detergent solution followed by clean water and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with copper must be decontaminated at the end of the laboratory work session, with all rinses collected as waste.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- POWDER spill of copper(II) nitrate: do NOT use paper towels, cellulose absorbent, or a broom on an oxidizer spill, and do not sweep it dry. If the spill is inside the operating fume hood and is less than approximately 1 g, trained staff may clean it up: dampen with water using a wash bottle, wipe up with a damp non-cellulose wipe or inert absorbent (vermiculite) working from the outside inward, and place the material in a screw-top waste container. Do not place oxidizer-contaminated absorbent in a container with organic waste.

- A powder spill outside the fume hood, or larger than approximately 1 g, or any spill contacting combustible material: restrict access, post a sign, remove any adjacent combustibles if it is safe to do so, and contact EH&S at 206-543-0467 during business hours or via 9-1-1 after hours.
- LIQUID spill - less than 250 mL of a dilute copper solution over secondary containment: may be handled by trained staff wearing goggles, nitrile gloves and a lab coat. Absorb with inert absorbent from the outside inward and place in a screw-top waste container.
- ANY copper-containing spill that reaches or could reach a sink, floor drain, or seawater line must be reported to EH&S immediately regardless of volume, because of Category 1 acute and chronic aquatic toxicity. Do not flush it away with water.
- Bead spill: pick up beads with a damp wipe or a magnet-free scoop; do not leave beads on the floor. If the beads were in a tube with a hazardous lysate, follow the spill procedure of the applicable SOP (usually SOP-05).
- PPE required for cleanup: chemical splash goggles, face shield for powder or concentrated solution, nitrile gloves, buttoned lab coat with disposable chemically resistant sleeves and apron, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - OXIDIZER SPILL' on the door for any powder spill outside the hood; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the area thoroughly with detergent solution followed by clean water and dry; confirm no blue-green residue remains.
- Spill cleanup materials must be disposed of in the following manner: place absorbent and contaminated disposables in a screw-top container SEPARATE from any organic or solvent waste, label with a UW Hazardous Waste Label listing 'copper(II) nitrate - oxidizer - spill debris', and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Eye exposure: use the eyewash for at least 15 minutes while holding the eyelids open, then go to the Emergency Department. Copper(II) nitrate causes serious eye damage.
- Skin exposure: wash the affected area with soap and running water for 15 minutes, using the safety shower for larger contact; remove contaminated clothing; use a clean lab coat or spare clothing for cover-up. Seek medical evaluation for any burn or persistent irritation.
- Inhalation exposure: move the person out of the contaminated area to fresh air; get medical help if coughing or irritation persists.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent, dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused copper(II) nitrate solid; (b) copper stock and working solutions; (c) copper exposure water and rinses; (d) copper-contaminated solid debris; (e) used zirconium oxide beads in lysate.
- NO copper-containing liquid may be poured down the drain, in any concentration, and none may be returned to a seawater system. Heavy metals are prohibited from sewer discharge and copper is acutely toxic to aquatic life at Category 1.
- Keep oxidizer waste SEPARATE from organic, solvent, and reducing-agent waste. Do not add copper(II) nitrate or its contaminated absorbent to a container holding solvent, DMSO, thiol or any organic waste. Use a dedicated inorganic/metal waste container.
- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible HDPE container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment, labeled 'COPPER - HEAVY METAL - OXIDIZER'.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Used zirconium oxide beads are inert but are contaminated by whatever lysate they were used with; dispose of the whole tube according to the SOP governing that lysate (usually SOP-05, phenol/guanidinium waste).
- No neutralization or in-lab treatment of copper waste is authorized. Copper cannot be neutralized; it can only be collected.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplused.

Section 7 - Protocol

- Procedure A - Weighing copper(II) nitrate and preparing stock solution. Inside the certified fume hood, with the balance in the hood and a plastic-backed absorbent liner in place, wearing goggles, nitrile gloves, lab coat and sleeves: weigh the salt into a disposable weigh boat with a disposable spatula (the salt is hygroscopic; work efficiently and recap the bottle immediately), transfer to a labeled vessel, add trace-metal-clean deionized water or dilute nitric or hydrochloric acid as the protocol specifies, and dissolve with stirring. Label with compound, concentration, 'OXIDIZER', 'NO DRAIN', date and initials. Remove and dispose of the liner promptly.
- Procedure B - Preparing copper exposure water. Dilute the stock into filtered seawater in a labeled, covered vessel standing in secondary containment. Verify the nominal concentration and, where the protocol requires it, submit a sample for analytical confirmation. Never prepare exposure water in a vessel connected to a flow-through system.
- Procedure C - Copper exposure experiments. Conduct exposures in static or static-renewal vessels within secondary containment in a posted area. Collect all used exposure water as hazardous waste at each renewal and at the end of the experiment. Do not return exposed animals or their water to a shared system.

- Procedure D - Bead-beating with zirconium oxide beads. Add beads to the tube at the bench, add tissue and lysis reagent inside the fume hood per the governing SOP, seal the tube, run the bead beater inside the fume hood, let the tube settle one minute, and open it inside the hood.
- Prohibited: heating copper(II) nitrate; storing it with combustibles or reducing agents; any drain or seawater discharge of copper; mixing oxidizer waste with organic waste; use by untrained personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

Yes, potentially. Copper exposure experiments with bivalves or other marine organisms are conducted under a current IACUC-approved protocol where IACUC oversight applies, and the approved protocol number is recorded here. Personnel handling exposed animals and their water must wear nitrile gloves, safety glasses and a lab coat; all exposure water, rinse water, exposed animal carcasses and cleaning materials are collected as chemical waste and are never discharged to a drain or seawater return; exposure vessels are kept in posted secondary containment; and tanks, nets and tubing used for copper exposures are labeled, dedicated to that purpose, and decontaminated per Section 10 before any other use, because residual copper can kill organisms in a later experiment.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with copper(II) nitrate must be trained on this SOP prior to starting work. They must also review the copper(II) nitrate SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDS; (4) documented instruction on oxidizer segregation and on the aquatic-release prohibition.
- A worker must complete at least one supervised run of Procedures A and B, observed and signed off by the PI or a designated experienced lab member, before preparing copper solutions independently.
- Copper exposure experiments involving live animals require a current IACUC-approved protocol, where applicable, and PI authorization before the exposure begins.
- Any new process that would heat copper(II) nitrate, combine it with an acid or organic material, or discharge any copper-containing liquid requires prior written PI approval and consultation with EH&S.
- No medical examination is required for the work described because respirators are not used.

Section 10 - Decontamination

- Fume hood and balance: remove and bag the absorbent liner, then wipe the work surface, sash interior and balance with a damp wipe, then detergent solution, then clean water; dry. Verify that no blue-green residue remains.

- Equipment: rinse pipettes, cylinders and stir bars three times into the copper waste container, then wash with detergent and water and rinse three times with deionized water. Glassware used for trace-metal work is additionally acid-washed per SOP-07 Procedure C, with all rinses collected.
- Exposure tanks, nets and tubing: drain into the copper waste container, wash with detergent and water, rinse three times with clean water collecting the first rinse as waste, and where practical dedicate them permanently to copper work rather than returning them to general use. Residual copper in shared equipment has killed subsequent experiments in marine laboratories.
- Bead beater: wipe the tube holder and chamber with detergent solution and 70% ethanol after each run; a failed tube requires decontamination inside the fume hood per the governing lysate SOP.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209; the designated areas in Section 11 are decontaminated at the end of each session.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, clean water, deionized water, dilute acid for trace-metal glassware, non-cellulose wipes and inert (vermiculite) absorbent for oxidizer residue.

Section 11 - Designated Area

- Weighing copper(II) nitrate and preparing stock solutions: the certified chemical fume hood in FTR 209, with the balance inside the hood and a plastic-backed absorbent liner in place. The hood is posted 'OXIDIZER IN USE - COPPER - NO DRAIN DISPOSAL' during this work.
- Copper exposure experiments: the posted exposure area in FTR 209 with secondary containment adequate for the full volume, labeled 'CONTAINS COPPER - TOXIC TO AQUATIC LIFE - NO DRAIN'.
- Bead-beating: the certified chemical fume hood in FTR 209.
- Waste containers for copper are kept within the designated areas in secondary containment, separated from organic and solvent waste containers.
- Storage: the oxidizers position of the chemical storage cabinet in FTR 209, in a non-combustible labeled secondary container, segregated from flammables, organics and reducing agents. Zirconium oxide beads are stored on the general reagent shelf.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.

I have read and understand the content of this SOP:

Name (print)	Signature	Date

[illegible]